Practice workbook · Explain how tooth decay happens and justify ways to prevent it

Sugar, bacteria and acid in plaque, why molars decay most, and brushing, fluoride, mouthwash, diet and dental visits, including decay data between regions or years.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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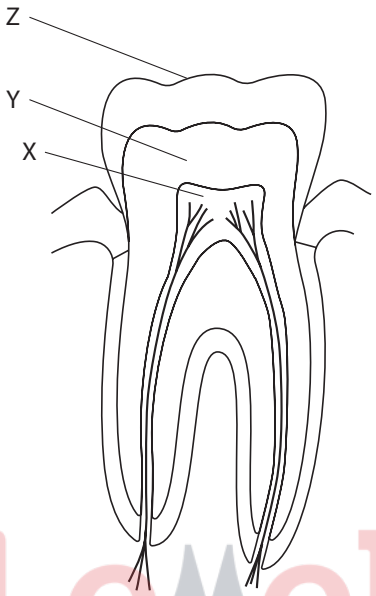
Type ID: HBI-TOOTH-DECAY-AND-CARE

1. Question 7 · part b · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b only.

7 The diagram shows a section through a tooth.



(a) (i) Name the parts of the tooth labelled X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

(3)

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1. Question 7 · part b · 4 marks

June 2023 Paper 2

Worked-example retry · Attempt part b only.

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(b) Explain the role of bacteria in the process of tooth decay.

(4)

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(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

(3)

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(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

(2)

(Total for Question 7 = 15 marks)

1. Question 7 · part b · 4 marks · Working and answer

Working and answer

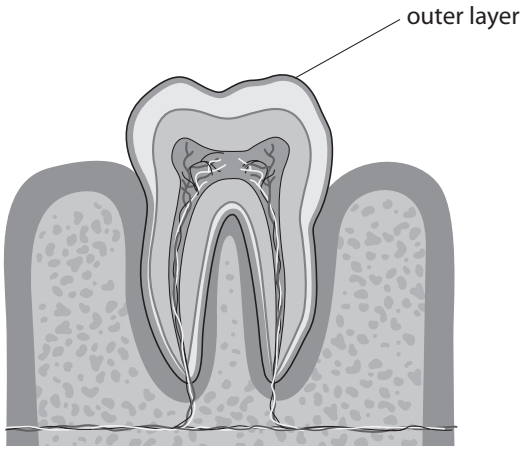
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2. Question 1 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

1 (a) The diagram shows a section through one type of tooth.



(i) State the name of this type of tooth.

(1)

.....

(ii) Which substance forms the outer layer of the tooth?

(1)

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

(3)

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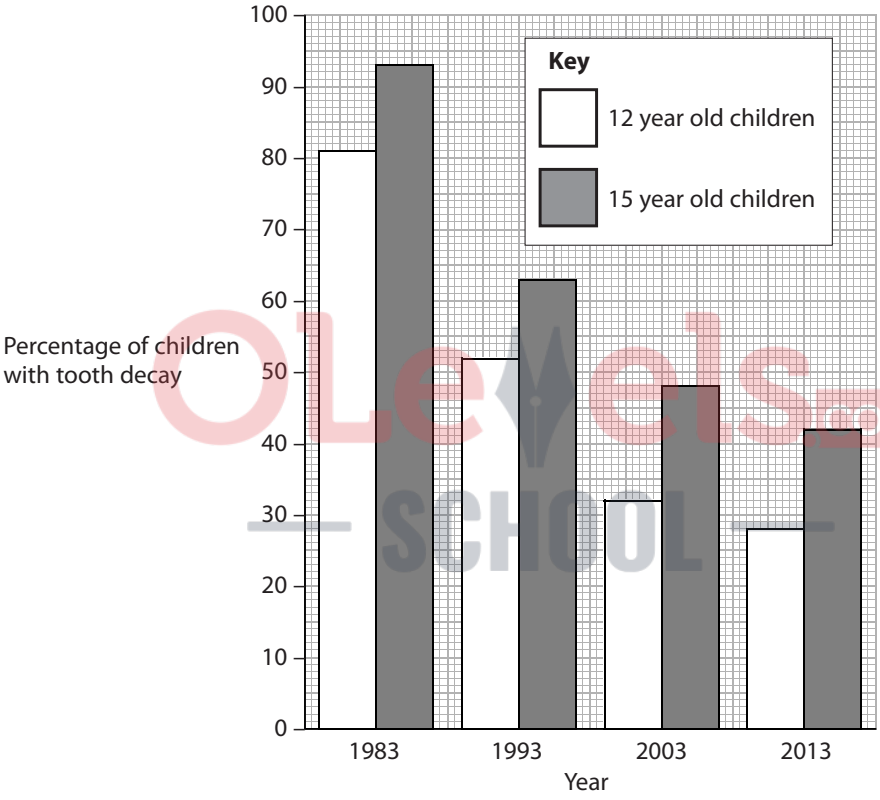
2. Question 1 · part a.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.iii only.

(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.



(i) Give two conclusions that can be made from the data in the bar chart.

(2)

1 .....

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2 .....

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(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

(1)

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(Total for Question 1 = 8 marks)

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2. Question 1 · part a.iii · 3 marks · Working and answer

Working and answer

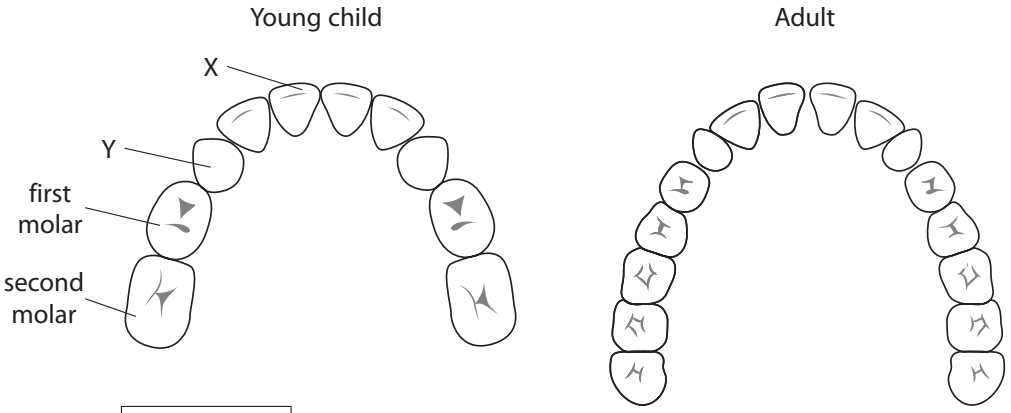
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3. Question 1 · part b.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

1 The diagram shows the upper teeth of a young child and the upper teeth of an adult.



not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

.....

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(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

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3. Question 1 · part b.ii · 2 marks

January 2020 Paper 2

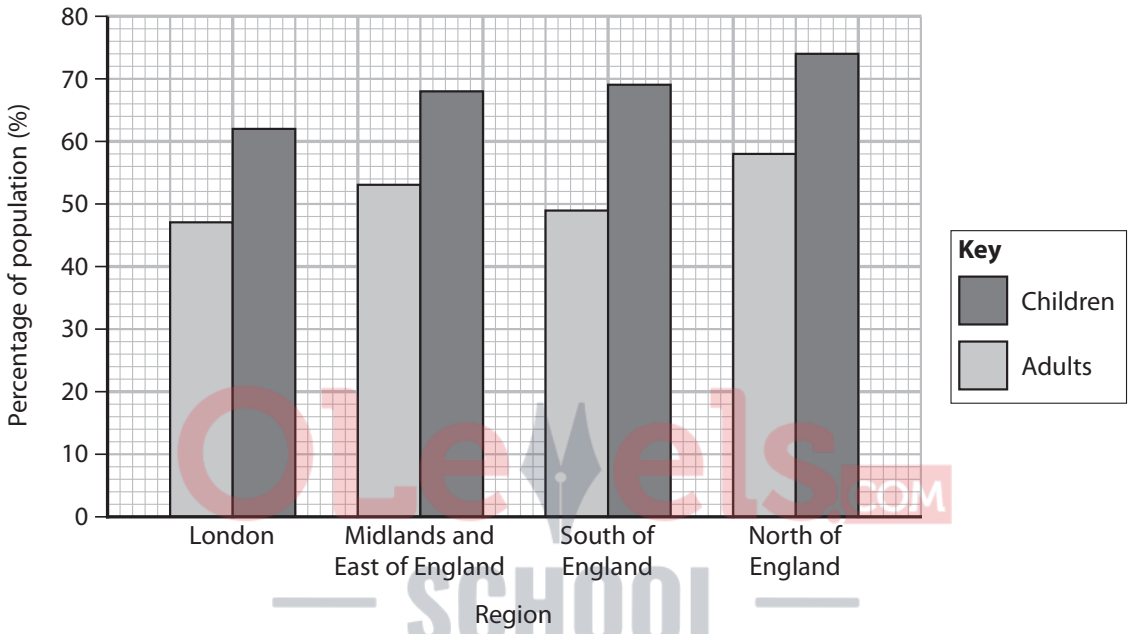
Further practice · Attempt part b.ii only.

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(b) The graph shows the percentage of adults and children in different regions of England who visited a dentist in a two-year period.



Data source/s: Health and Social Care Information Centre NHS Dental Activity Statistics: December 2013 and GP Patient Survey Dental Results 2013/14

(i) Explain which region is likely to have more adults and children with tooth decay.

(2)

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(ii) Give two reasons why molars are more likely to be damaged by tooth decay than other teeth.

(2)

1 .....

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2 .....

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(Total for Question 1 = 9 marks)

3. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part c.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part c.i only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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4. Question 4 · part c.i · 4 marks

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Further practice · Attempt part c.i only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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4. Question 4 · part c.i · 4 marks

November 2021 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Explain how reducing the amount of sugar in foods helps prevent tooth decay.

(4)

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(ii) State two other methods that could reduce tooth decay.

(2)

1 .....

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(Total for Question 4 = 15 marks)

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4. Question 4 · part c.i · 4 marks · Working and answer

Working and answer

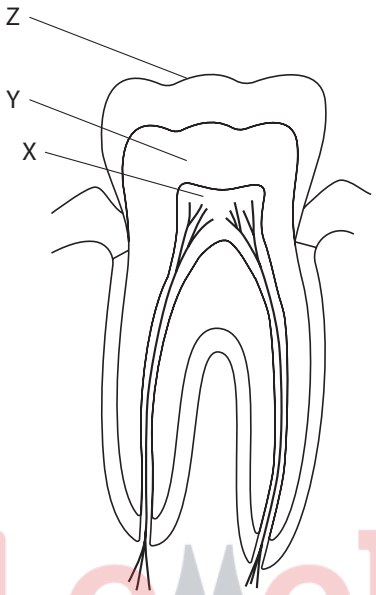
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5. Question 7 · part a.ii · 3 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

7 The diagram shows a section through a tooth.



(a) (i) Name the parts of the tooth labelled X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

(3)

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5. Question 7 · part a.ii · 3 marks

June 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) Explain the role of bacteria in the process of tooth decay.

(4)

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

(3)

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

(2)

(Total for Question 7 = 15 marks)

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5. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

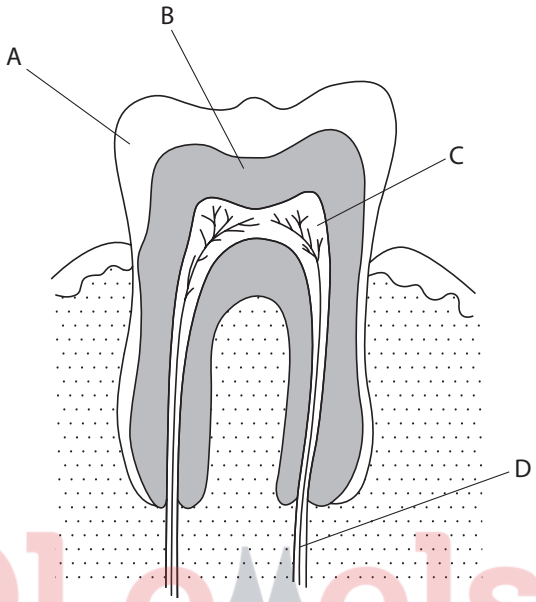
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6. Question 3 · part a.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

3 (a) The diagram shows a tooth.



(i) Name the type of tooth.

(1)

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(ii) Identify the parts of the tooth labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

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6. Question 3 · part a.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.iii only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

(3)

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

(4)

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(c) State three other ways to keep teeth healthy.

(3)

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(Total for Question 3 = 15 marks)

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6. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

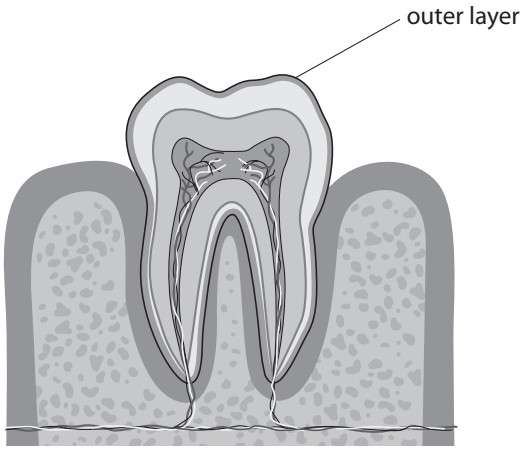
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7. Question 1 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a section through one type of tooth.



(i) State the name of this type of tooth.

(1)

(ii) Which substance forms the outer layer of the tooth?

(1)

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

(3)

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7. Question 1 · part b.ii · 1 mark

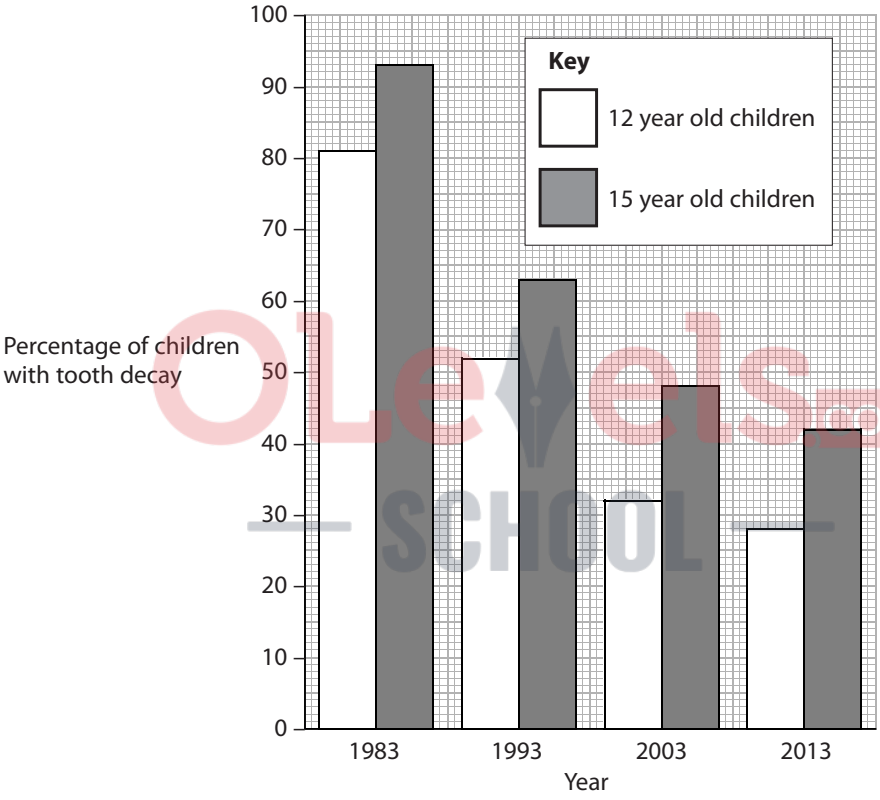
June 2019 Paper 2

Further practice · Attempt part b.ii only.

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(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.



(i) Give two conclusions that can be made from the data in the bar chart.

(2)

1 .....

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2 .....

.....

(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

(1)

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(Total for Question 1 = 8 marks)

7. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

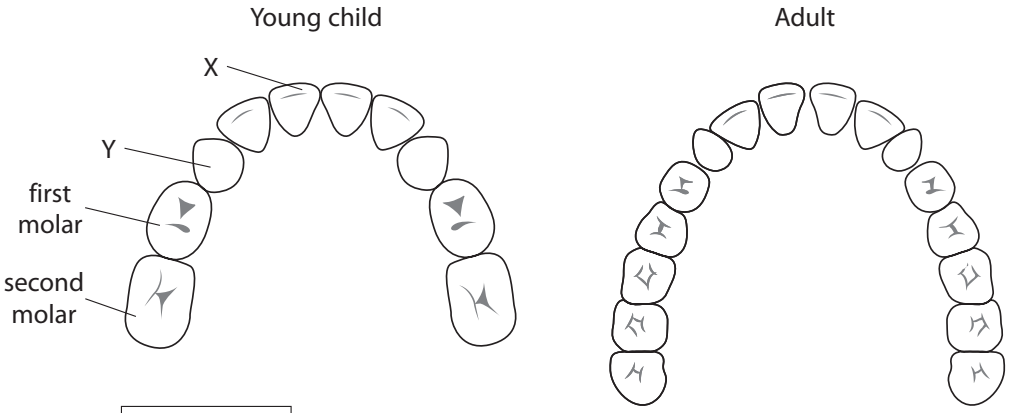
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8. Question 1 · part b.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part b.i only.

1 The diagram shows the upper teeth of a young child and the upper teeth of an adult.



not to scale

(a) (i) Name tooth X and tooth Y. **(2)**

tooth X .....

tooth Y .....

(ii) State the function of the molar teeth. **(1)**

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(iii) State two differences between the teeth of a young child and the teeth of an adult. **(2)**

1 .....

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2 .....

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8. Question 1 · part b.i · 2 marks

January 2020 Paper 2

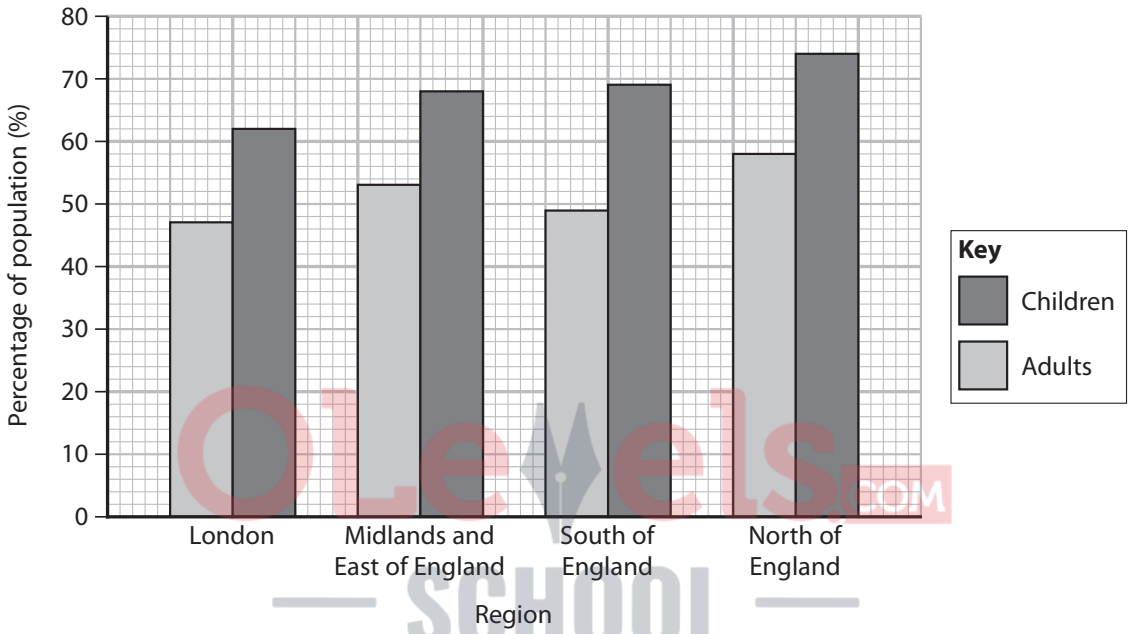
Further practice · Attempt part b.i only.

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(b) The graph shows the percentage of adults and children in different regions of England who visited a dentist in a two-year period.



Data source/s: Health and Social Care Information Centre NHS Dental Activity Statistics: December 2013 and GP Patient Survey Dental Results 2013/14

(i) Explain which region is likely to have more adults and children with tooth decay.

(2)

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(ii) Give two reasons why molars are more likely to be damaged by tooth decay than other teeth.

(2)

1 .....

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2 .....

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(Total for Question 1 = 9 marks)

8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

4 The table shows the results of a survey about the number of teeth per student in a class of young students.

Number of teeth	Number of students
23	1
24	2
25	3
26	4
27	7
28	13

(a) (i) Draw a bar chart to show the results of the survey.

(4)



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9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

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(ii) A full set of teeth for the students is 28

Calculate the percentage of the students in the class who have a full set of teeth.

(3)

percentage = ... %

(b) A full set of teeth for an adult is 32

Suggest why the full set of teeth for these students is only 28

(2)

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9. Question 4 · part c.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Explain how reducing the amount of sugar in foods helps prevent tooth decay.

(4)

(ii) State two other methods that could reduce tooth decay.

(2)

1 .....

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(Total for Question 4 = 15 marks)

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9. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

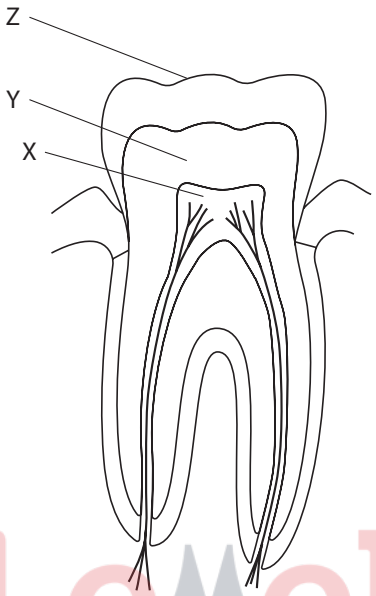
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10. Question 7 · part d · 2 marks

June 2023 Paper 2

Further practice · Attempt part d only.

7 The diagram shows a section through a tooth.



(a) (i) Name the parts of the tooth labelled X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Explain where decay in the tooth is most likely to occur.

(3)

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10. Question 7 · part d · 2 marks

June 2023 Paper 2

Further practice · Attempt part d only.

(b) Explain the role of bacteria in the process of tooth decay.

(4)

(c) Babies are usually born without teeth. The teeth develop many months after birth.

Explain why the teeth do not develop until many months after birth.

(3)

(d) Dentists recommend using a mouthwash morning and night to reduce dental decay.

One type of mouthwash contains alcohol and sodium fluoride.

Explain how this mouthwash could help to reduce dental decay.

(2)

(Total for Question 7 = 15 marks)

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10. Question 7 · part d · 2 marks · Working and answer

Working and answer

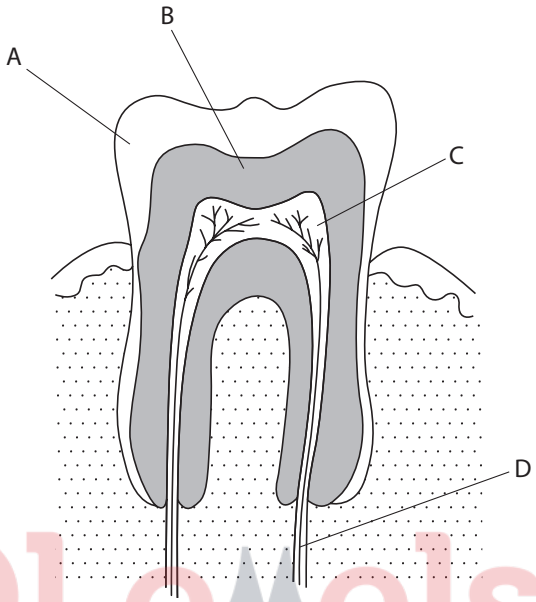
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11. Question 3 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

3 (a) The diagram shows a tooth.



(i) Name the type of tooth.

(1)

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(ii) Identify the parts of the tooth labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

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11. Question 3 · part b · 4 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

(3)

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

(4)

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(c) State three other ways to keep teeth healthy.

(3)

1 .....

2 .....

3 .....

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(Total for Question 3 = 15 marks)

11. Question 3 · part b · 4 marks · Working and answer

Working and answer

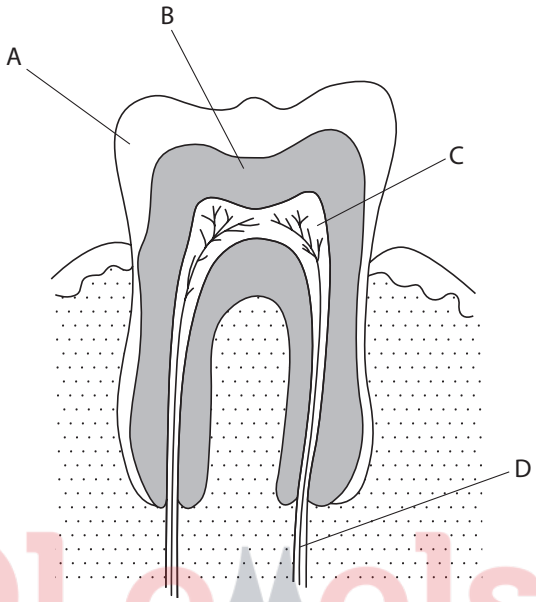
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12. Question 3 · part c · 3 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

3 (a) The diagram shows a tooth.



(i) Name the type of tooth.

(1)

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(ii) Identify the parts of the tooth labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

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12. Question 3 · part c · 3 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

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(iii) Explain why this type of tooth is more likely to have tooth decay than other types of teeth.

(3)

(b) Explain why the minerals obtained from a balanced diet are important in developing and maintaining healthy teeth.

(4)

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(c) State three other ways to keep teeth healthy.

(3)

1 .....

2 .....

3 .....

(Total for Question 3 = 15 marks)

12. Question 3 · part c · 3 marks · Working and answer

Working and answer

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